

The algebraic properties of real numbers

Axiom (Field Axioms for $\mathbb{R}$): The algebraic properties of $\mathbb{R}$ can be derived from:

- Commutativity: $x+y = y+x$ and $x \times y = y \times x$, for all $x, y \in \mathbb{R}$.
- Associativity: $(x+y)+z = x+(y+z)$ and $(x \times y) \times z = x \times (y \times z)$, for all $x, y, z \in \mathbb{R}$.
- Identity: there exists the elements $0, 1 \in \mathbb{R}$ such that, for every $x \in \mathbb{R}$, $0+x = x = x+0$ and $1 \times x = x = x \times 1$.
- Inverses: for every $x \in \mathbb{R}$, there exists $w, u \in \mathbb{R}$ such that $x+w = 0 = w+x$ and $x \times u = 1 = u \times x$.
- Distributivity: $x \times (y+z) = x \times y + x \times z$, for all $x, y, z \in \mathbb{R}$.

Axiom (Order properties of $\mathbb{R}$): These are

- Trichotomy: for every $x \in \mathbb{R}$, exactly one of the following holds: $x \in P$, $x=0$, $-x \in P$.
- Compatibility with $+$ and $\times$: $(x+y), (x \times y) \in P$, for every $x, y \in P$.